1 **(a)** State the name of the class (group) of fundamental particles that contains a neutrino.

.... [1]

(b) A hadron P has a charge of +1*e*, where *e* is the elementary charge. The hadron P is composed of a down antiquark and only one other quark.

(i) Identify a possible flavour for this other quark.

.... [1]

(ii) State what type of hadron is P.

.... [1]

(c) Nucleus Q undergoes radioactive decay to form nucleus R, emitting an antineutrino and another particle X, as shown in the decay equation.

$$Q \longrightarrow R + X + \overline{\nu}$$

(i) State what particle is represented by X.

.... [1]

(ii) Compare the nucleon numbers of Q and R.

.... [1]

(iii) Compare the charges of Q and R.

.... [1]

2 A nucleus P undergoes α-decay to form nucleus Q.

(a) Complete the equation for this decay.

$$^{215}_{84}P \longrightarrow {}^{\_\_}_{\_\_}Q + {}^{\_\_}_{\_\_}\alpha$$ [2]

(b) **(i)** State the principle of conservation of momentum.

....

....

.... [2]

(ii) Before the decay, nucleus P has a speed of $3.2 \times 10^5\,m\,s^{-1}$. After the decay, nucleus Q is stationary.

Calculate the speed of the alpha particle after the decay.

speed = ... $m\,s^{-1}$ [2]

[Total: 6]

3 Nuclei of an isotope of copper (Cu) each have 29 protons and 37 neutrons. This isotope is a β^- emitter.

(a) State the nuclide notation in the form $^{A}_{Z}X$ for this nucleus of copper.

[1]

(b) The energy spectrum of the β^- radiation emitted by a sample of this isotope is shown in Fig. 7.1.

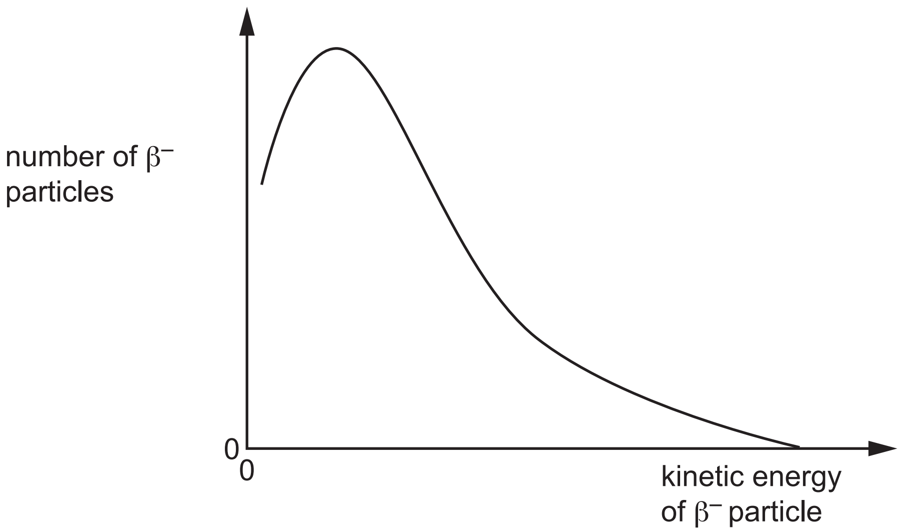

Fig. 7.1

(i) Use Fig. 7.1 to explain why other particles apart from the β^- particles must be emitted during this decay.

....

....

....

....

.... [3]

(ii) State the name of the other particle emitted during the decay of this isotope.

.... [1]

(iii) The copper isotope decays to an isotope of zinc (Zn).

Give the radioactive decay equation for this decay. Include the nucleon and proton numbers of **all** the particles involved.

[3]

[Total: 8]

4 Nuclei of an isotope of samarium (Sm) each contain 62 protons and 85 neutrons.

(a) State the nuclide notation in the form $^{A}_{Z}X$ for this isotope of samarium.

[1]

(b) This isotope of samarium is radioactive and decays by emitting particles. Gamma-radiation is **not** emitted. The energy spectrum of the emitted particles is shown in Fig. 6.1.

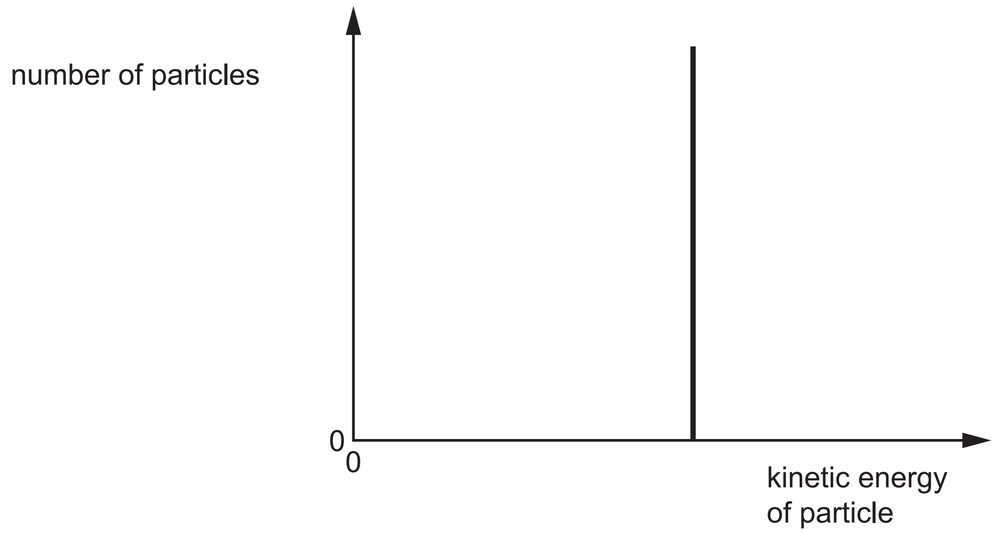

Fig. 6.1

(i) Explain how Fig. 6.1 shows that this isotope of samarium emits α-particles and does **not** emit β-particles.

....

....

....

.... [2]

(ii) This isotope of samarium decays to an isotope of neodymium (Nd).

Give the radioactive decay equation for this decay. Include the nucleon and proton numbers of **all** the particles involved.

[2]

(c) A baryon is composed of three quarks which all have different flavours. The baryon has a charge of 0.

Two of the quarks in the baryon are an up quark and a bottom quark.

(i) Determine, in terms of the elementary charge *e*, the charge on the third quark in the baryon.

charge = ... *e* [2]

(ii) State a possible flavour for the third quark in the baryon.

.... [1]

[Total: 8]

5 **(a)** Potassium-40 ($^{40}_{19}K$) undergoes β^- decay to form a nuclide of element X. Particle Z is emitted during the decay. The equation for the decay is shown.

$$^{40}_{19}K \rightarrow \, ^{P}_{Q}X + \, ^{R}_{S}\beta^- + Z$$

(i) State the values of *P*, *Q*, *R* and *S*.

P = ... *R* = ...

Q = ... *S* = ...

[2]

(ii) State the name of particle Z.

.... [1]

(iii) State the name of the class of fundamental particle to which both the β^- particle and particle Z belong.

.... [1]

(b) Determine the quark composition of an alpha-particle.

quark composition .... [3]

[Total: 7]

6 **(a)** Compare an α-particle with a β^+ particle in terms of their masses and charges.

....

....

....

....

.... [3]

(b) Nucleus P undergoes α-decay to form nucleus Q. Nucleus Q then undergoes a further decay to form nucleus R. The proton and nucleon numbers of P and R are shown in Fig. 6.1.

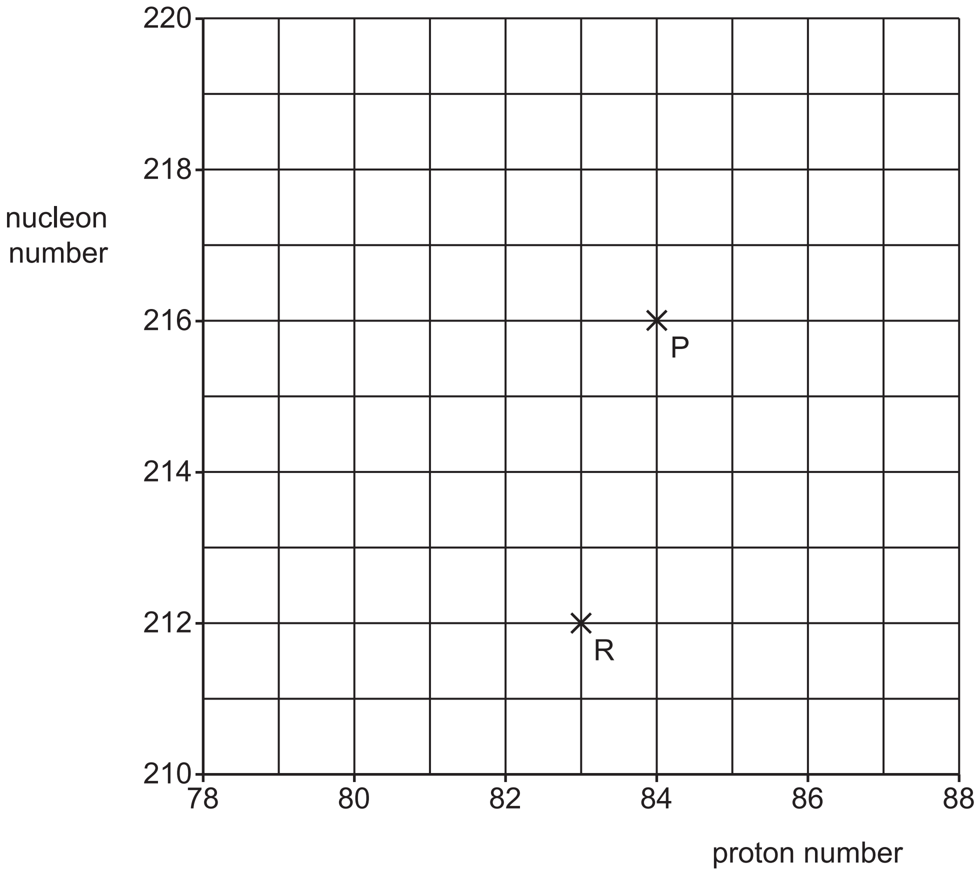

Fig. 6.1

(i) On Fig. 6.1, draw a cross (×) to show the proton number and nucleon number of Q. Label your cross Q. [1]

(ii) State the names of the particles emitted as Q decays to form R.

....

.... [2]

(c) Before the α-decay, P is travelling at a constant velocity. After the decay, Q has a velocity of $1.3 \times 10^5\,\mathrm{m\,s^{-1}}$ at an angle of 68° to the original path of P.
The α-particle has a velocity of $150 \times 10^5\,\mathrm{m\,s^{-1}}$ at an angle of θ to the original path of P, as shown in Fig. 6.2.

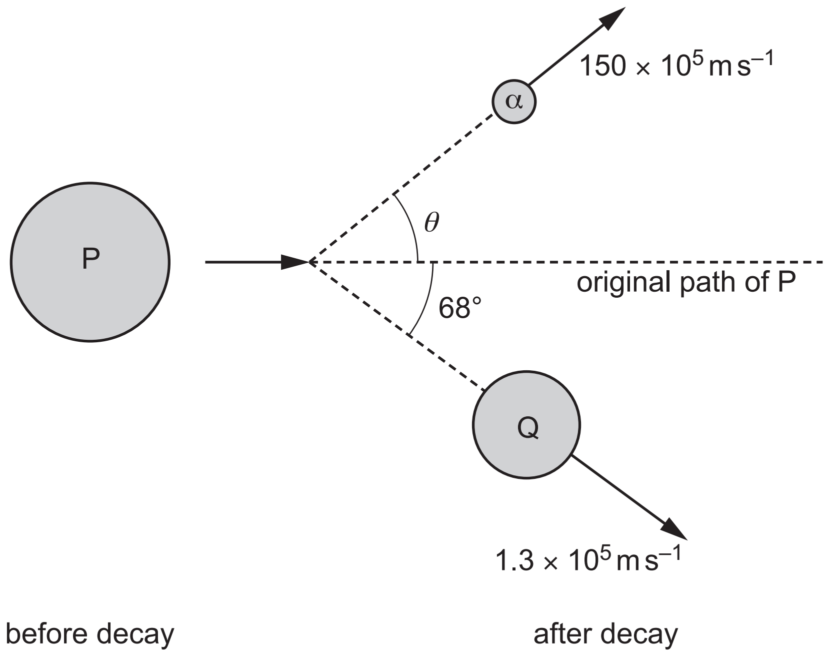

Fig. 6.2 (not to scale)

(i) Use the principle of conservation of momentum to determine θ.

θ = ...° [3]

(ii) Calculate the kinetic energy of the α-particle.

kinetic energy = ... J [2]

[Total: 11]

7 **(a)** Complete Table 7.1 to show the charges, in terms of the elementary charge *e*, on each of the flavours of quark and antiquark shown.

Table 7.1

flavour	charge / *e*	
	quark	antiquark
up		
down		
strange		

[3]

(b) (i) State the name of the class (group) of fundamental particles to which baryons and mesons belong.

.... [1]

(ii) Compare baryons and mesons in terms of their constituent particles.

....

....

.... [2]

(c) Describe β^+ decay in terms of the fundamental particles involved.

....

....

....

.... [2]

8 A particle Q and a particle R are each composed of one quark and one antiquark.

(a) State the name of the class (group) of particles that includes Q and R.

.... [1]

(b) Q has a charge of –1*e*, where *e* is the elementary charge. R has a charge of 0.

Complete Table 7.1 to show a possible second quark in each of Q and R.

Table 7.1

	charge	first quark	second quark
Q	–1*e*	strange	
R	0	anti-up	

[2]

[Total: 3]